Graduate Project 2026: Sales Data Processing and Business Insights System

1. Project Overview

This project simulates a real-world data analysis task where graduates must design and implement a **data processing and reporting workflow** using Linux, Python, and Excel.

Graduates are required to:

- Process and clean raw data using Python
- Automate execution using Linux
- Perform analysis using Excel Pivot Tables
- Present clear, actionable business insights

---

2. Project Objective

Develop a system that transforms raw transactional data into structured insights to support business decision-making.

---

3. Business Scenario

You are a Data Analyst working for a retail organization. The business collects transactional sales data but lacks visibility into performance.

Management requires answers to the following:

- Which regions are performing best?
- Which products generate the highest revenue?
- How do sales trends change over time?
- Which salespeople are top performers?

You are responsible for building a solution that processes the data and delivers these insights.

---

4. Technical Requirements

4.1 Linux Environment

Graduates must:

- Set up a structured project directory:

```
project/
```

```
├── data/
├── scripts/
├── output/
└── reports/
```

- Use Linux commands to:
  - Navigate directories
  - Manage files
  - Execute scripts
- Create a shell script:

```
run_pipeline.sh
```

  This script must:
  - Execute the Python processing script
  - Save outputs to the appropriate directory
  - Log execution status

4.2 Python Data Processing

Create a Python script:

```
process_data.py
```

The script must:

Data Ingestion

- Load raw CSV data from the data directory

Data Cleaning

- Handle missing values
- Remove duplicates
- Standardize formats (dates, text, numeric fields)

Data Transformation

- Create new calculated fields:
  - Revenue = Quantity × Price
  - Month extracted from date

Data Aggregation

Generate summary datasets:

- Sales by region

- Sales by product
- Monthly revenue
- Salesperson performance

Output

Export:

- Clean dataset → `output/clean_sales.csv`
- Aggregated datasets → separate CSV files

4.3 Excel Analysis (Pivot Tables)

Using the cleaned dataset, graduates must:

Create Pivot Tables:

- Sales by region
- Top-performing products
- Monthly revenue trends
- Salesperson performance

Create Visualizations:

- Bar chart: Sales by region
- Line chart: Monthly revenue
- Pie chart: Product category distribution

Dashboard Requirement:

- Organize all visuals into a single, clean Excel dashboard

4.4 Business Analysis and Reporting

Graduates must produce a concise report addressing:

1. Top-performing region
2. Highest revenue-generating products
3. Sales trends over time
4. Top-performing salespeople
5. At least three actionable business recommendations

---

5. Deliverables

Each graduate must submit:

1. Python script (`process_data.py`)
2. Linux shell script (`run_pipeline.sh`)
3. Cleaned dataset and aggregated outputs
4. Excel file containing Pivot Tables and dashboard
5. Written report (1–2 pages)
6. Short presentation (5–10 minutes)

---

6. Timeline (1 Week)

Day	Activity
Day 1	Environment setup and dataset familiarization
Day 2	Python data cleaning
Day 3	Data transformation and aggregation
Day 4	Excel Pivot Tables
Day 5	Dashboard creation
Day 6	Business analysis and report writing
Day 7	Final review and presentation

---

7. Evaluation Criteria

Category	Weight
Data Processing (Python)	25%
Automation (Linux)	15%
Data Analysis (Excel)	25%
Business Insights	20%

Presentation and Documentation	15%

---

8. Expected Competencies

Upon completion, graduates should demonstrate the ability to:

- Work within a Linux-based environment
- Develop structured Python scripts for data processing
- Transform raw data into analysis-ready datasets
- Use Excel Pivot Tables for efficient data summarization
- Translate data findings into business insights
- Communicate results clearly and professionally

---

9. Extension (Optional for High Performers)

Graduates may extend the project by:

- Adding logging to the Python script
- Automating Excel file generation using Python
- Introducing error handling and validation
- Creating a more advanced dashboard with slicers

---

10. Outcome

This project ensures that graduates can:

- Integrate multiple tools into a cohesive workflow
- Apply technical skills in a business context
- Demonstrate independent problem-solving ability

Data set:

Sales_data.csv
09 Apr 2026, 04:01 PM